

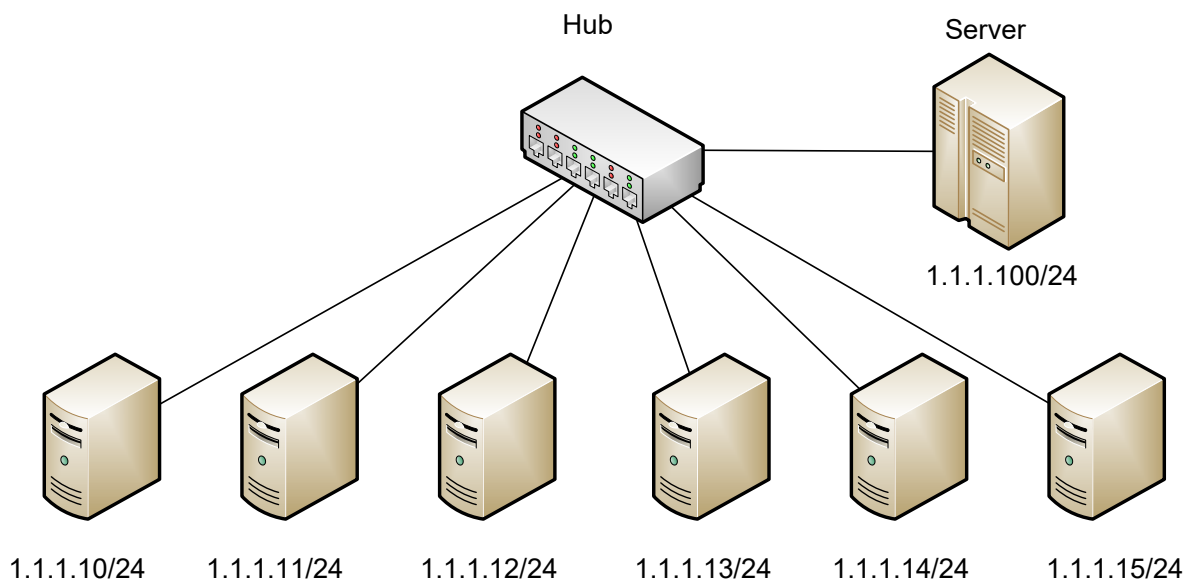
Lab 6

DHCP and DNS Services

Objective:

The purpose of this lab is for the student to observe how DHCP and DNS services function. Students will configure a server for DHCP and DNS functionality.

Diagram:



Components:

- 6 – PC's with an Ethernet NIC
- 1 – Hub
- 1 – Server (with DHCP and DNS ability)

Procedure:

1. Connect all PC's and the server together and put their static IP addresses on them. Verify connectivity by pinging each machine.
2. Load the Server 2008 Virtual Machine in VMWare workstation. The file should be in C:\Capal Install. There will be a separate video about VMWare workstation.
3. Install the DHCP service on the Server. This step might already be done from previous semesters. If not, Go to Start, Control Panel, Administrative Tools, Server Manager. Expand and Click on Roles. Click Add Roles and follow the wizard by selecting the DHCP role. Click Install and Finish the wizard.
4. Click Start, Administrative Tools, DHCP to load the DHCP Manager.
5. Create a new scope named Network1
6. Make the IP ranges 1.1.1.50 – 1.1.1.70 with a subnet mask of /24
7. Make the lease duration 1 day.
8. You do not need to configure DNS or gateway options for this lab so you can skip them.
9. Once completed, activate the scope
10. Change the IP addresses on the PC's from static to dynamic (select obtain IP address automatically). **Do not change the IP settings on the server, it must remain static.**
11. Once the setting has been changed disable and re-enable the network card.

12. Go to the command prompt and view the advanced ip address information.

- a. What is the new IP address? 1.1.1.52
- b. What is the MAC address? D8-9E-F3-49-AC-3B
- c. Does the command show the lease time? Yes
- d. Does the command show the DHCP server IP? Yes

13. Run the ipconfig /release command, then run the ipconfig /all command.

- a. What IP address do you see? 169.254.87.122(Tentative)
- b. Are the lease times shown? No

14. Run the ipconfig /renew command, then run the ipconfig /all command.

- a. What IP address do you see? 1.1.1.52(Preferred)
- b. Are the lease times shown? Yes
- c. Is this the same IP address as before or a different one? Why?
Yes, because the DHCP server assigned the MAC address this IP address with a lease ending tomorrow, thus when the same MAC address

15. Go to the DHCP server and view the address leases.

- a. What information is shown?
Client IP address, Client name, Lease Expiration for all of the clients in our pod.

16. Disable the DHCP scope on the server.

17. Open up the command prompt on the PC's and type in ipconfig /release and then ipconfig /renew?

- a. Is the time for the command to execute affected? Why?
Yes, because the scope of the DHCP Server was disabled; the server is not responding to the request because the scope was disabled, so the host is probably looking for a new DHCP server to configure its IP.

18. Once the command finished, view the IP address. What is the IP address now? Why?

The IP address is now 169.254.87.122(Preferred). This is because the host is not connected to a DHCP server, so the host auto-assigned itself an IP from the 169.254.x.x

19. Go to the server and activate the scope again.

20. Run the ipconfig /renew command on the PC's.

- a. Is the time for the command to execute affected? Why? Yes, it took longer than before.
- b. What IP address did you receive?
1.1.1.52

21. Go to the server and setup a reservation for your PC. Use the IP address of 1.1.1.80.

22. Run the ipconfig /renew command on the PC's.

- a. What IP address did you receive?
1.1.1.80

23. Install the DNS service on the server. This might already be done from previous semesters. If not, Go to Start, Control Panel, Administrative Tools, Server Manager. Expand and Click on Roles. Click Add Roles and

follow the wizard by selecting the DNS role. Click Install and Finish the wizard.

24. Click Start, Administrative Tools, DNS to launch the DNS Manager.

25. Within the DNS service, create a new DNS zone called class.com with all of the default options.

26. Create a new name record of class.com with the IP address of the server. Create name records PC-A, PC-B, and PC-C. With the IP addresses 1.1.1.10, 1.1.1.11, and 1.1.1.12 respectively.

27. On the PC's change the DNS settings to include the IP address of the DNS server.

28. Open a command prompt and ping class.com. Was the ping successful?
No, but we were able to ping each individual (PC-A.class.com)

29. Ping -A, PC-B, and PC-C. Were the pings successful?
Yes they were.

30. Ping PC-D. Was the ping successful?
No, because this PC does not exist.

31. Go to the server and create a new name record of PC-D with an IP address of 1.1.1.13. Go back to the PC and try to ping PC-D. Was the ping successful?
Yes

32. Run the command ipconfig /flushdns. Ping PC-D. Was the ping successful? Why?

Yes, it was. I suspect that the DNS flush would cause the ping to be slower or less successful, but it did not; mine operated as normal even after the flushing.

33. Draw the lab diagram below (PC's and hub connections) Label the IP addresses, and subnet masks of each device.

